Exact two-soliton S-matrix candidates and the constraints on them for the good and bad Boussinesq equations

Project *quantum_boussinesq_solitons* (Phase 2, next step 5)

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Abstract

We classify candidate exact quantum two-soliton S-matrices $S(1,2;\hbar)$ for both the *good* (fixed convention) and *bad* Boussinesq equations across overtaking and head-on channels, using only constraints that are independent of the pending Toda-to-continuum quantum frame dictionary: (i) the semiclassical limit $\hbar \arg S \to \delta(1,2)$, including $O(\hbar^0)$ Maslov/Stirling offsets; (ii) unitarity; (iii) quantum crossing; (iv) bound-state poles and zeros against the physical breather spectrum; and (v) consistency with the N-soliton asymptotic Bethe equations on a ring. Keeping the good and bad equations strictly separate with explicit dictionaries, we establish four main results **(c)**:

1. **Exclusion of mixed-argument $\mathfrak{sl}_3$ quantum dilogarithms and all-orders mirror products.** Because $\partial_1\partial_2(\ell_1 - \ell_2) = 0$, the mixed derivative $\partial_1\partial_2\delta(1,2)$ is a rational function of $(e^{\beta_1}, e^{\beta_2})$ (or $(e^{i\alpha_1}, e^{i\alpha_2})$) of transcendental weight zero. By contrast, for any quantum dilogarithm $\Phi_\hbar(R)$ with tree-level phase $\mathrm{Li}_2(R)$, $\partial_1\partial_2 \mathrm{Li}_2(R)$ retains a transcendental $\ln(1-R)$ term unless $\partial_1\partial_2 \ln R = 0$ (excluding mixed arguments such as $R = A_{12}$), while multiplicatively separable KP root ratios $R_{rs} = \tan(\varphi_{1,r}/2)/\tan(\varphi_{2,s}/2)$ fail at the diagonal pole $\alpha_1 \to \alpha_2$ (relative least-squares residual 3.68×10^{-2}). Mirror products $F(D)G(S)$ are excluded at *all* orders in $\hbar$ by two-body exchange unitarity $S(1,2)S(2,1) = 1$ alone, which forces $G(S) = \text{const}$.
2. **Good vs. bad bound-state poles.** For the bad equation, the Gamma candidate S_Γ^{bad} has simple physical-sheet poles precisely at the Bogdan–Charkina breather levels $I_{\mathrm{br}} = N\hbar$ ($N \geq 1$) and no pole at $N = 0$. For the good equation, every complex-conjugate Hirota breather has $A_{12} < 1/4$ and is singular **(a)**/**(c)**; nevertheless, on the light branch $\alpha_{1,2} = a \mp ib$ with $a \in (0, \pi/3)$ and $b < 0$, $\mathrm{Im}(P_1 - P_2) > 0$ and $i\eta < 0$, so the naive continuation S_Γ^{good} acquires spurious simple poles on the physical sheet at singular kinematics ($A_{12} < 1/4$). Across all real Y-junction resonances ($A_{12} = 0$ or ∞), $\ell_1 = \ell_2$ ($\eta = 0$) and $S_\Gamma^{\mathrm{good}} \to e^{iT_{12}/\hbar}$ smoothly without poles.
3. **Twisted N-soliton ring Bethe equations from topological cross terms.** Transporting soliton j around a ring of circumference Λ across the seam shifts its Darboux coordinate by $\Delta_{\mathrm{seam}} X_j' = \Lambda + \gamma_j \sum_{l \neq j} M_l$ and drops the seam potential $\phi(0^+)$ by M_j, producing the Aharonov–Bohm ϕ-winding twist $+\Pi_j M_l$. The resulting twisted directional Bethe phase $\Phi_{\mathrm{ring}}(j \mid l) = \delta(j,l) + \Pi_j M_l$ satisfies $-\partial_{P_j}\Phi_{\mathrm{ring}}(j \mid l) = \Delta x_{jl} = -\frac{1}{k_j}\ln A_{jl}$ (the exact Hirota shift), obeys the global translation-winding sum rule $P_{\mathrm{tot}}\Lambda + \Pi_{\mathrm{tot}} M_{\mathrm{tot}} = 2\pi\hbar \sum_j n_j$, and fuses covariantly onto Bogdan–Charkina breathers.
4. **Minimal residual ambiguity and package interfaces.** After (i)–(v), the residual freedom at $O(\hbar^0)$ reduces to a single real antisymmetric overtaking phase $\vartheta_{\mathrm{ov}}(1,2) = -\vartheta_{\mathrm{ov}}(2,1)$, with $\vartheta_{\mathrm{ho}}(\beta_1,\beta_2) = -\vartheta_{\mathrm{ov}}(\beta_1, \beta_2 + i\pi)$ fixed by crossing. We prove the exact crossing identity $I_{12}^{\mathrm{same}}(\alpha_1, \alpha_2 + \pi) = -I_{12}^{\mathrm{head}}(\alpha_1, \alpha_2)$ and $J_{12}^{\mathrm{same}}(\alpha_1, \alpha_2 + \pi) + J_{12}^{\mathrm{head}}(\alpha_1, \alpha_2) = +\pi/2$ for the one-loop continuum integrals, showing that the $+\pi/2$ cusp jump of J_{12} across $\alpha_2 = \pi/3$ cancels the $+\pi/2$ Stirling offset of $S_{\Gamma,\mathrm{ho}}$, and state the exact interface formulas to the projected determinant (`det_*`) and quantum frame (`frame_*`) packages.

Every candidate passing tests remains a guess **(d)** until derived from a quantum theory.

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1 Conventions, good/bad dictionaries, and scope

1.1 Separate conventions for the good and bad equations

Throughout this note we keep the *good* Boussinesq equation (the fixed repository convention of `AGENTS.md` and `review.pdf`) and the *bad* Boussinesq equation (used in the Toda and Bogdan–Charkina analyses of `report.pdf` and `smatrix_ansatz.pdf`) strictly separate.

Good equation (fixed convention). The potential $\phi(x,t)$, velocity field $u=\phi_x=2(\ln\tau)_{xx}$, Lagrangian density L, and canonical symplectic form Ω are

$$u_{tt}=u_{xx}-u_{xxxx}-3(u^2)_{xx},\qquad L=\tfrac{1}{2}\phi_t^2-\tfrac{1}{2}\phi_x^2+\phi_x^3-\tfrac{1}{2}\phi_{xx}^2,\qquad \Omega=\int_{-\infty}^{\infty}d\pi\wedge d\phi\quad(\pi=\phi_t),\tag{1}$$

and $\hbar$ refers to this canonical normalisation. A one-soliton solution has wavenumber $k\in(0,1)$, direction $\epsilon\in\{+1,-1\}$, and rapidities uniformised by $\alpha\in(0,\pi/2)$:

$$k=\sin\alpha,\quad v=\epsilon\cos\alpha,\quad \omega=kv=\tfrac{1}{2}\epsilon\sin 2\alpha,\quad \ell=-\epsilon\cos 3\alpha,\quad M=2k,\quad \Pi=-Mv=-\epsilon\sin 2\alpha,\tag{2}$$

with canonical momentum and energy **(a)**/**(c)** (`review.pdf`)

$$P=\tfrac{2}{3}\epsilon k^3\cos\alpha,\qquad E=\tfrac{2}{15}k^3(5-4k^2),\qquad \frac{dP}{d\alpha}=\tfrac{2}{3}\epsilon\sin\alpha\sin 3\alpha=\frac{2k}{9}\frac{d\ell}{d\alpha}.\tag{3}$$

Bad equation. For the bad equation $v_{tt} = v_{xx} + v_{xxxx} + 3(v^2)_{xx}$ with $v = -\tilde{\phi}_x = -2(\ln\tau)_{xx}$ and $L_{\rm bad} = \frac{1}{2}\tilde{\phi}_t^2 - \frac{1}{2}\tilde{\phi}_x^2 + \tilde{\phi}_x^3 + \frac{1}{2}\tilde{\phi}_{xx}^2$, a one-soliton solution is supersonic ($|v| > 1$) and uniformised by $\beta > 0$:

$$k = \sinh\beta, \quad v = \epsilon\cosh\beta, \quad \omega = kv = \tfrac{1}{2}\epsilon\sinh 2\beta, \quad \ell = \epsilon\cosh 3\beta, \quad M = 2k, \quad \Pi = -Mv = -\epsilon\sinh 2\beta, \tag{4}$$

with canonical momentum and energy **(a)**/**(c)** (`report.pdf`)

$$P = \tfrac{2}{3}\epsilon k^3\cosh\beta, \qquad E = \tfrac{2}{15}k^3(5+4k^2), \qquad \frac{dP}{d\beta} = \tfrac{2}{3}\epsilon\sinh\beta\sinh 3\beta = \frac{2k}{9}\frac{d\ell}{d\beta}. \tag{5}$$

Table 1 records the complete dictionary between the two equations, including the analytic continuation $\alpha = i\beta$ (with the sign flip in ℓ: $\ell_{\rm good}(i\beta) = -\epsilon\cos(3i\beta) = -\epsilon\cosh 3\beta = -\ell_{\rm bad}(\beta)$).

Table 1: Explicit dictionary between the good (fixed convention) and bad Boussinesq equations **(c)** (`scripts/smx_dictionary_and_families.py`).

Quantity	Good equation (fixed convention)	Bad equation
PDE	$u_{tt} = u_{xx} - u_{xxxx} - 3(u^2)_{xx}$	$v_{tt} = v_{xx} + v_{xxxx} + 3(v^2)_{xx}$
Potential / Tau	$u = \phi_x = 2(\ln\tau)_{xx}$	$v = -\tilde{\phi}_x = -2(\ln\tau)_{xx}$
Uniformisation	$\alpha \in (0, \pi/2)$	$\beta \in (0, \infty)$
Wavenumber, speed	$k = \sin\alpha,\ v = \epsilon\cos\alpha\ \ (\lvert v\rvert < 1)$	$k = \sinh\beta,\ v = \epsilon\cosh\beta\ \ (\lvert v\rvert > 1)$
Spectral parameter ℓ	$\ell = -\epsilon\cos 3\alpha$	$\ell = +\epsilon\cosh 3\beta$
Mass, shift charge	$M = 2\sin\alpha,\ \Pi = -Mv = -\epsilon\sin 2\alpha$	$M = 2\sinh\beta,\ \Pi = -Mv = -\epsilon\sinh 2\beta$
Momentum, energy	$P = \frac{2}{3}\epsilon k^3\cos\alpha,\ E = \frac{2}{15}k^3(5-4k^2)$	$P = \frac{2}{3}\epsilon k^3\cosh\beta,\ E = \frac{2}{15}k^3(5+4k^2)$
Darboux ratio $\gamma = d\Pi/dP$	$\gamma = -\dfrac{3\cos 2\alpha}{\sin\alpha\sin 3\alpha}$	$\gamma = -\dfrac{3\cosh 2\beta}{\sinh\beta\sinh 3\beta}$
Hirota polynomial $\mathcal{P}(K, W)$	$W^2 - K^2 + K^4$	$W^2 - K^2 - K^4$
KP cubic roots (p_a, p_b)	$\frac{1}{2}(\pm k + v/\sqrt{3}),\ \ 4p^3 - p = \frac{\ell}{3\sqrt{3}}$	$\frac{1}{2}(\pm k + iv/\sqrt{3}),\ \ 4p^3 + p = \frac{i\ell}{3\sqrt{3}}$
Overtaking $A_{12}^{\rm ov}$ $(\epsilon_1 = \epsilon_2)$	$\dfrac{1-3\cot^2(S/2)}{1-3\cot^2(D/2)} = \dfrac{G_{\rm good}(S/2)}{G_{\rm good}(D/2)}$	$\dfrac{1+3\coth^2(S/2)}{1+3\coth^2(D/2)} = \dfrac{G_{\rm bad}(S/2)}{G_{\rm bad}(D/2)}$
Head-on $A_{12}^{\rm ho}$ $(\epsilon_1 = -\epsilon_2)$	$\dfrac{1-3\tan^2(D/2)}{1-3\tan^2(S/2)} = \dfrac{1}{A_{12}^{\rm ov}(\alpha_1, \alpha_2+\pi)}$	$\dfrac{1+3\tanh^2(D/2)}{1+3\tanh^2(S/2)} = \dfrac{1}{A_{12}^{\rm ov}(\beta_1, \beta_2+i\pi)}$
$\mathfrak{sl}_3$ block $G(y)$	$G_{\rm good}(y) = \dfrac{4\cos^2 y - 1}{\sin^2 y}$	$G_{\rm bad}(y) = \dfrac{4\cosh^2 y - 1}{\sinh^2 y}$
Crossing involution	$\alpha_2 \mapsto \alpha_2 + \pi\ \ (k_2 \mapsto -k_2,\ v_2 \mapsto -v_2)$	$\beta_2 \mapsto \beta_2 + i\pi\ \ (k_2 \mapsto -k_2,\ v_2 \mapsto -v_2)$

1.2 Two-soliton semiclassical phase and exact log-derivative identity

In both equations, let particle 1 start to the left of particle 2 as $t \to -\infty$ ($x_1 < x_2$ for $\epsilon_1 = \epsilon_2 = +1$ with $v_1 > v_2$, or for head-on $\epsilon_1 = +1, \epsilon_2 = -1$). The classical position shifts are $\Delta x_{12} = -\frac{1}{k_1}\ln A_{12}$ and $\Delta x_{21} = +\frac{1}{k_2}\ln A_{12}$, while the Darboux-coordinate shifts are $\Delta x'_{12} = \Delta x_{12} - \gamma_1 M_2$ and $\Delta x'_{21} = \Delta x_{21} + \gamma_2 M_1$ (`report.pdf`, Sec. 3.2). Integrating $-\Delta x'_{12}\,dP_1 - \Delta x'_{21}\,dP_2 = d\delta(1,2)$ yields the semiclassical phase **(c)**:

$$\delta(1,2) = \delta_{\rm dyn}(1,2) + T_{12}, \qquad \delta_{\rm dyn}(1,2) = \frac{2}{9}(\ell_1 - \ell_2)\ln A_{12}, \qquad T_{12} = \frac{2}{3}M_1M_2(v_1 - v_2) = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1). \tag{6}$$

The fact that $\partial_1\delta(1,2) = -P'_1\Delta x'_{12}$ holds identically in both equations rests on a remarkable algebraic cancellation between the derivative of $\ln A_{12}$ and the topological term:

Proposition 1.1 (**(c)**, `scripts/smx_dictionary_and_families.py`)**.** *For both the good equation (with $\partial_1 = \partial_{\alpha_1}$) and the bad equation (with $\partial_1 = \partial_{\beta_1}$), and in both overtaking ($\epsilon_1 = \epsilon_2$) and head-on ($\epsilon_1 = -\epsilon_2$) channels, the interaction constant A_{12} satisfies the KP cross-ratio identity*

$$A_{12} = \frac{(p_{1a} - p_{2a})(p_{1b} - p_{2b})}{(p_{1a} - p_{2b})(p_{1b} - p_{2a})}, \tag{7}$$

and the exact log-derivative identity

$$\frac{2}{9}(\ell_1-\ell_2)\,\partial_1 \ln A_{12} = -\frac{1}{3}\Pi_1' M_2 - \frac{2}{3} M_1' \Pi_2. \tag{8}$$

Consequently, differentiating $\delta(1,2) = \frac{2}{9}(\ell_1-\ell_2)\ln A_{12} - \frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1)$ *with respect to the rapidity of soliton 1 gives*

$$\partial_1 \delta(1,2) = \frac{2}{9}\ell_1' \ln A_{12} - \Pi_1' M_2 = P_1'\Big(\frac{1}{k_1}\ln A_{12} - \gamma_1 M_2\Big) = -P_1' \Delta x_{12}', \tag{9}$$

where we used $\frac{2}{9}\ell_1' = P_1'/k_1$ *from* (3) *and* (5). *Verified to 35 digits at six test pairs across both equations and channels (residual* $< 10^{-35}$*).*

1.3 Scope of this write-up and deferred comparisons

A separate work package (`frame_*`) is concurrently deriving the quantum momentum and frame dictionary between the quantum Toda chain at finite density and the continuum bad Boussinesq equation, while another (`det_*`) is auditing the projected one-loop determinant. Accordingly:

- **Deferred quantitative Toda edge comparisons:** Every quantitative $O(\hbar^0)$ comparison with numerical Toda Fermi-edge data (`toda_edge.pdf`, `matching.pdf`), including the conjectured small-amplitude overtaking residual $c_0^*(r) = -\frac{4}{\pi}(\chi_2(r) - r)$ in Eq. (19) of `report.pdf` and the finite-$\hbar_T$ reciprocity remainder of `audit_toda_dictionary.pdf`, is **explicitly deferred** to a subsequent run once the `frame_*` dictionary is completed.
- **Exclusion of Phase 2i:** The note `phase2/toda_wh_and_kp_synthesis.tex` (Phase 2i) is under independent audit (`audit_phase2i*`) and is *not* relied upon anywhere in this document.

2 Classification of candidate S-matrix families

We test candidate two-body phase functions $S(1,2;\hbar)$ in both overtaking ($\epsilon_1 = \epsilon_2$) and head-on ($\epsilon_1 = -\epsilon_2$) channels against five structural requirements:

(i) **Semiclassical limit:** $\hbar \arg S(1,2;\hbar) \to \delta(1,2)$ as $\hbar \to 0$ at fixed rapidities in both channels, tracking the $O(\hbar^0)$ offset $\lim_{\hbar\to 0}\big(\arg S - \delta/\hbar\big)$;

(ii) **Unitarity:** $|S(1,2;\hbar)| = 1$ on the regular real rapidity manifold and two-body exchange unitarity $S(1,2;\hbar)S(2,1;\hbar) = 1$;

(iii) **Quantum crossing:**

$$S_{\rm ov}(\beta_1,\beta_2{+}i\pi;\hbar)\,S_{\rm ho}(\beta_1,\beta_2;\hbar) = 1 \quad \text{(bad)}, \qquad S_{\rm ov}(\alpha_1,\alpha_2{+}\pi;\hbar)\,S_{\rm ho}(\alpha_1,\alpha_2;\hbar) = 1 \quad \text{(good)}; \tag{10}$$

(iv) **Bound-state poles and zeros:** Poles on the physical sheet $\mathrm{Im}(P_1 - P_2) > 0$ must match the physical bound-state spectrum (Sec. 3);

(v) **Ring Bethe consistency:** Compatibility with the N-soliton twisted asymptotic Bethe equations on a ring (Sec. 4).

2.1 Family I: Gamma ansatz, string shifts s, and η-CDD modifications

Let $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ and define the $\hbar$-independent regularised cross-ratios

$$\mathcal{A}_{12} = \frac{81 A_{12}}{4(\ell_1 - \ell_2)^2}, \qquad \mathcal{B}_{12} = \frac{81}{4 A_{12}(\ell_1 - \ell_2)^2}, \tag{11}$$

so that $A_{12}/\eta^2 = \hbar^2 \mathcal{A}_{12}$ and $1/(A_{12}\eta^2) = \hbar^2 \mathcal{B}_{12}$. Consider the shifted Gamma family with parameter $s > 0$ **(d)**:

$$S_{\mathrm{ov}}^{(s)}(1,2) = \frac{\Gamma(s + i\eta)}{\Gamma(s - i\eta)} \big(e^2 \hbar^2 \mathcal{A}_{12}\big)^{i\eta} e^{iT_{12}/\hbar} \qquad (\epsilon_1 = \epsilon_2), \tag{12}$$

$$S_{\mathrm{ho}}^{(s)}(1,2) = \frac{\Gamma(s - i\eta)}{\Gamma(s + i\eta)} \big(e^2 \hbar^2 \mathcal{B}_{12}\big)^{-i\eta} e^{iT_{12}/\hbar} \qquad (\epsilon_1 = +1,\ \epsilon_2 = -1), \tag{13}$$

and $S_{-+}^{(s)}(1,2) = 1/S_{+-}^{(s)}(2,1)$, together with rational or hyperbolic CDD factors in η:

$$F_{\mathrm{rat}}(\eta; c) = \frac{c + i\eta}{c - i\eta} \quad (c \in \mathbb{R} \setminus \{0\}), \qquad F_{\mathrm{hyp}}(\eta; \gamma) = \frac{\sinh \frac{\pi}{2}(\eta - i\gamma)}{\sinh \frac{\pi}{2}(\eta + i\gamma)} \quad (\gamma \in \mathbb{R}). \tag{14}$$

Proposition 2.1 (**(c)**, `scripts/smx_dictionary_and_families.py`). *1. For every $s > 0$, the family (12)–(13) satisfies unitarity (ii) and exact crossing (iii) (whenever $\mathcal{A}_{12}, \mathcal{B}_{12} > 0$), and has the Stirling asymptotic expansion as $\hbar \to 0$ ($|\eta| \to \infty$) at fixed rapidities:*

$$\arg S^{(s)}(1,2) - \frac{\delta(1,2)}{\hbar} = \Big(s - \frac{1}{2}\Big)\pi\tau\,\mathrm{sgn}(\eta) - \frac{s^2 - s + \frac{1}{6}}{\eta}\tau + O(\eta^{-3}) \pmod{2\pi}, \tag{15}$$

where $\tau = +1$ for overtaking ($\epsilon_1 = \epsilon_2$) and $\tau = -1$ for head-on ($\epsilon_1 = +1, \epsilon_2 = -1$). In particular:

- *For the baseline candidate $s = 1$ (`smatrix_ansatz.pdf`), the $O(\hbar^0)$ Maslov/Stirling offset is $+\frac{\pi}{2}\,\mathrm{sgn}(\eta)$ in overtaking and $-\frac{\pi}{2}\,\mathrm{sgn}(\eta)$ in head-on, and the $O(\hbar^1)$ coefficient is $-\frac{1}{6\eta}\tau$.*
- *For the half-integer shift $s = \frac{1}{2}$, the $O(\hbar^0)$ constant offset* vanishes *in both channels, and the $O(\hbar^1)$ coefficient is $+\frac{1}{12\eta}\tau$, while the poles shift to $i\eta = -(N - \frac{1}{2})$ ($N \geq 1$).*

2. Multiplying $S_{\mathrm{ov}}^{(s)}$ by $F_{\mathrm{CDD}}(\eta)$ and $S_{\mathrm{ho}}^{(s)}$ by $F_{\mathrm{CDD}}(\eta)^{-1}$ preserves unitarity (ii) and crossing (iii). As $\hbar \to 0$ at fixed rapidities ($|\eta| \to \infty$), any η-only CDD factor (14) contributes only a rapidity-independent constant step *at $O(\hbar^0)$:*

$$\arg F_{\mathrm{rat}}(\eta; c) = \pi\,\mathrm{sgn}(c)\,\mathrm{sgn}(\eta) - \frac{2c}{\eta} + O(\eta^{-3}), \qquad \arg F_{\mathrm{hyp}}(\eta; \gamma) = -\pi\gamma\,\mathrm{sgn}(\eta) + O(e^{-\pi|\eta|}). \tag{16}$$

Thus η-only CDD factors cannot generate a smooth rapidity-dependent $O(\hbar^0)$ phase at fixed rapidities; conversely, any nontrivial rapidity-dependent $O(\hbar^0)$ phase $\vartheta(\beta_1, \beta_2)$ must depend directly on (β_1, β_2) rather than through η alone.

2.2 Family II: $\hbar$-renormalisation and coupling shifts

In quantum integrable field theories such as sine-Gordon and affine Toda, the bare coupling $\hbar$ is renormalised in the dynamical soliton phase by $\hbar \mapsto \hbar_{\mathrm{ren}} = \hbar(1 + c_1\hbar + c_2\hbar^2 + \dots)$ **(a)** [5, 7]. In the Boussinesq candidate (12)–(13), $\hbar$ enters in two distinct places: in the spectral ratio $\eta = \frac{2}{9}(\ell_1 - \ell_2)/\hbar$ (which controls the dynamical phase δ_{dyn} and the breather pole spacing), and in the topological Aharonov–Bohm factor $e^{iT_{12}/\hbar}$.

Proposition 2.2 (**(c)**, `scripts/smx_dictionary_and_families.py`). *1. The dynamical phase* $\delta_{\rm dyn}(1,2) = \frac{2}{9}(\ell_1 - \ell_2)\ln A_{12}$ *and the topological term* $T_{12} = \frac{2}{3}M_1M_2(v_1 - v_2)$ *obey the classical crossing symmetry* separately*:*

$$\delta^{\rm ov}_{\rm dyn}(\beta_1, \beta_2 + i\pi) = -\delta^{\rm ho}_{\rm dyn}(\beta_1, \beta_2), \qquad T^{\rm ov}_{12}(\beta_1, \beta_2 + i\pi) = -T^{\rm ho}_{12}(\beta_1, \beta_2), \tag{17}$$

and identically under $\alpha_2 \mapsto \alpha_2 + \pi$ *in the good equation (verified to 35 digits).*

2. *Because of* (17)*, both* global $\hbar$*-renormalisation* ($\hbar \mapsto \hbar_{\rm ren}$ *in both* η *and* $e^{iT_{12}/\hbar}$) *and* dynamical-only $\hbar$*-renormalisation* ($\eta \mapsto \eta_{\rm ren} = \frac{2}{9}(\ell_1 - \ell_2)/\hbar_{\rm ren}$ *while keeping canonical* $\hbar$ *in* $e^{iT_{12}/\hbar}$) *preserve exact unitarity (ii) and exact quantum crossing (iii).*

3. *Expanding as* $\hbar \to 0$ *at fixed rapidities,* $\hbar_{\rm ren} = \hbar(1 + c_1\hbar + c_2\hbar^2 + \dots)$ *shifts the* $O(\hbar^0)$ *phase by a term proportional to the classical phase:*

$$\arg S_{\rm ren}(1,2) - \arg S_0(1,2) = \begin{cases} -c_1\,\delta(1,2) + O(\hbar) & (\textit{global } \hbar \mapsto \hbar_{\rm ren}), \\ -c_1\,\delta_{\rm dyn}(1,2) + O(\hbar) & (\textit{dynamical-only } \eta \mapsto \eta_{\rm ren}), \end{cases} \tag{18}$$

and shifts the Bogdan–Charkina breather quantisation condition to $I_{\rm br} = N\hbar_{\rm ren} = N\hbar(1 + c_1\hbar + O(\hbar^2))$.

2.3 Family III: $\mathfrak{sl}_3$ quantum-dilogarithm candidates (separated and mixed arguments)

In `smatrix_ansatz.pdf` (Sec. 2, p. 2), quantum dilogarithms $\Phi_\hbar$ were excluded at leading order only when their arguments are functions of $D = \beta_1 - \beta_2$ and $S = \beta_1 + \beta_2$ separately; whether a product of quantum dilogarithms with *mixed* two-soliton arguments (such as A_{12}, KP root differences $\mu_{1a} - \mu_{2b}$, or KP root ratios) could reproduce $\delta(1,2)/\hbar$ at leading order was left as an open question. We now resolve this question completely.

Recall that any Faddeev/compact quantum dilogarithm $\Phi_\hbar(R)$ with semi-classical expansion $\ln\Phi_\hbar(R) = \frac{1}{i\hbar}\,\mathrm{Li}_2(R) + O(\hbar^0)$ contributes a tree-level phase spanned by Euler dilogarithms $\mathrm{Li}_2(R_k(\beta_1, \beta_2))$. Since single-soliton phase conventions vanish under the mixed derivative $\partial_1\partial_2$, any candidate built from quantum dilogarithms (times $e^{iT_{12}/\hbar}$) must match $\partial_1\partial_2\delta_{\rm dyn}(1,2)$ via a linear combination $\sum_k c_k\,\partial_1\partial_2\,\mathrm{Li}_2(R_k)$.

Theorem 2.3 (**(c)**, `scripts/smx_dictionary_and_families.py`). *1.* ***Transcendental weight zero of*** $\partial_1\partial_2\delta(1,2)$*.* *Because* $\ell_1 - \ell_2$ *is a sum of single-particle functions,* $\partial_1\partial_2(\ell_1 - \ell_2) = 0$*. Hence*

$$\partial_1\partial_2\delta_{\rm dyn}(1,2) = \frac{2}{9}\Big[\ell_1'\,\partial_2\ln A_{12} - \ell_2'\,\partial_1\ln A_{12} + (\ell_1 - \ell_2)\,\partial_1\partial_2\ln A_{12}\Big] \tag{19}$$

is a purely rational function of $(e^{\beta_1}, e^{\beta_2})$ *(bad equation) or* $(e^{i\alpha_1}, e^{i\alpha_2})$ *(good equation), containing* no logarithms or polylogarithms *(transcendental weight zero; verified symbolically in SymPy).*

2. ***Exclusion of non-separable mixed arguments.*** *For any twice-differentiable argument* $R(\beta_1, \beta_2)$*,*

$$\partial_1\partial_2\,\mathrm{Li}_2\big(R(\beta_1,\beta_2)\big) = \frac{R}{1-R}(\partial_1\ln R)(\partial_2\ln R) - \ln\big(1 - R(\beta_1,\beta_2)\big)\,\partial_1\partial_2\ln R. \tag{20}$$

Therefore, $\partial_1\partial_2\,\mathrm{Li}_2(R)$ *has transcendental weight zero if and only if* $\partial_1\partial_2\ln R = 0$*, i.e.,* $R(\beta_1, \beta_2) = f_1(\beta_1)/f_2(\beta_2)$ *is multiplicatively separable. In particular, mixed arguments such as* $R = A_{12}(\beta_1, \beta_2)$ *(for which* $\partial_1\partial_2\ln A_{12} = 5.4541932 \neq 0$ *at* $(\beta_1, \beta_2) = (1.1, 0.4)$*) or additive root differences* $R = c(\mu_{1a} - \mu_{2b})$ *retain a transcendental* $\ln(1-R)$ *term and cannot match* (19)*.*

3. ***Exclusion of multiplicatively separable KP root-ratio dilogarithms at*** $O(\hbar^{-1})$**.** *In the KP uniformisation where the three roots of soliton* j *are* $\mu_{j,r} = \frac{1}{\sqrt{3}} \sin \varphi_{j,r}$ *with* $\varphi_{j,r} = \alpha_j + \frac{2\pi r}{3}$ *(*$r \in \{0,1,2\}$*), the natural multiplicatively separable ratios are* $R_{rs}(\alpha_1, \alpha_2) = \tan(\varphi_{1,r}/2)/\tan(\varphi_{2,s}/2)$*, which satisfy* $\partial_{\alpha_1}\partial_{\alpha_2} \ln R_{rs} = 0$ *and*

$$\partial_{\alpha_1}\partial_{\alpha_2} \operatorname{Li}_2(R_{rs}) = -\frac{R_{rs}}{1 - R_{rs}} \frac{1}{\sin \varphi_{1,r} \sin \varphi_{2,s}}. \tag{21}$$

On 40 *random regular overtaking pairs* (α_1, α_2)*, the* 10*-parameter basis* $\operatorname{span}_{r,s}\{\partial_1\partial_2 \operatorname{Li}_2(R_{rs})\} \oplus \operatorname{span}\{\partial_1\partial_2 T_{12}\}$ *fails least-squares matching to* $\partial_1\partial_2\delta(1,2)$ *with relative residual* 3.68×10^{-2}*. Analytically, as* $\alpha_1 \to \alpha_2 = \alpha$*, the diagonal* $(r = s)$ *term in* (21) *has residue* $-\frac{1}{2\sin\varphi_r}\frac{1}{\alpha_1 - \alpha_2}$*, whose numerator is bounded as* $\sin\varphi_r \to 0$*, whereas* $\partial_1\partial_2\delta_{\mathrm{dyn}}(1,2)$ *has diagonal pole residue* $+\frac{4}{3}\frac{\sin 3\alpha}{\alpha_1 - \alpha_2}$ *(good) or* $+\frac{4}{3}\frac{\sinh 3\beta}{\beta_1 - \beta_2}$ *(bad).*

Remark 2.4 (**(c)**). While Theorem 2.3 excludes quantum dilogarithms as the *leading* $O(\hbar^{-1})$ tree-level building block, classical dilogarithms $\operatorname{Li}_2(R_{rs})$ of precisely the KP root ratios $R_{rs} = \tan(\varphi_{1,r}/2)/\tan(\varphi_{2,s}/2)$ appear at $O(\hbar^0)$ (one loop) in the continuum integral $I_{12}(\alpha_1, \alpha_2)$ (`oneloop_phase.pdf`; see Sec. 5).

2.4 Family IV: Mirror products $F(\beta_1 - \beta_2)G(\beta_1 + \beta_2)$

Because the $\mathfrak{sl}_3$ factorisation $A_{12}^{\mathrm{ov}} = G(S/2)/G(D/2)$ splits $\ln A_{12}$ into a function of $S = \beta_1 + \beta_2$ minus a function of $D = \beta_1 - \beta_2$, Route C of `report.pdf` asked whether the S-matrix could be a "mirror" product $M(\beta_1, \beta_2) = F(\beta_1 - \beta_2)G(\beta_1 + \beta_2)$ (possibly times $e^{iT_{12}/\hbar}$). Phase 2c excluded this at $O(\hbar^{-1})$ because $\partial_S\partial_D\delta_{\mathrm{dyn}} \neq 0$. We observe that exchange unitarity (ii) excludes any nontrivial $G(\beta_1 + \beta_2)$ factor at *all* orders in $\hbar$:

Proposition 2.5 (**(c)**, `scripts/smx_dictionary_and_families.py`)**.** 1. *At* $O(\hbar^{-1})$*,* $\partial_S\partial_D\delta_{\mathrm{dyn}} \neq 0$ *in both equations:* $\partial_S\partial_D\delta_{\mathrm{dyn}}^{\mathrm{bad}} = -3.0850$ *at* $(S, D) = (1.4, 0.6)$ *and* $\partial_S\partial_D\delta_{\mathrm{dyn}}^{\mathrm{good}} = -1.6310$ *at* $(S, D) = (0.8, 0.3)$*.*

2. *At* any *order in* $\hbar$*, let* $M(\beta_1, \beta_2; \hbar) = F(\beta_1 - \beta_2; \hbar)G(\beta_1 + \beta_2; \hbar)$ *be a multiplicative block in an overtaking S-matrix satisfying two-body exchange unitarity* $M(\beta_1, \beta_2; \hbar)M(\beta_2, \beta_1; \hbar) = 1$*. Because exchanging* $1 \leftrightarrow 2$ *sends* $D \mapsto -D$ *and* $S \mapsto +S$*,*

$$F(D;\hbar)F(-D;\hbar)\,G(S;\hbar)^2 = 1 \quad \Longrightarrow \quad \frac{\partial}{\partial S}\ln\big(G(S;\hbar)^2\big) = 0 \quad \Longrightarrow \quad G(S;\hbar) = \textit{const.} \tag{22}$$

Thus no non-constant factor $G(\beta_1 + \beta_2; \hbar)$ *can appear multiplicatively in* $S(1, 2; \hbar)$*; the* $\beta_1 + \beta_2$ *dependence of* A_{12} *can enter* $S(1, 2; \hbar)$ *only inside an exponent multiplied by an antisymmetric prefactor such as* $i\eta(1, 2) = -i\eta(2, 1)$*.*

2.5 Family V: Pure and continuum-dressed exponentials

Consider the pole-free exponential family **(d)**

$$S_{\exp}(1, 2; \hbar) = \exp\Big(\frac{i}{\hbar}\delta(1, 2) + i\delta_1(1, 2) + O(\hbar)\Big), \tag{23}$$

where $\delta_1(1, 2)$ is a real one-loop phase obeying antisymmetry $\delta_1(1, 2) + \delta_1(2, 1) = 0$ and crossing $\delta_1^{\mathrm{ov}}(\beta_1, \beta_2 + i\pi) + \delta_1^{\mathrm{ho}}(\beta_1, \beta_2) = 0 \pmod{2\pi}$.

- For the **bad equation**, $S_{\exp}$ passes (i)–(iii) and (v), but is **excluded by (iv)**: it is an entire function of η with no poles on the physical sheet, failing to reproduce the Bogdan–Charkina breather spectrum $I_{\mathrm{br}} = N\hbar$ **(a)**/**(c)** [1].

- For the **good equation**, where regular classical breathers do not exist (Sec. 3), $S_{\rm exp}^{\rm good}$ is *not* excluded by (iv); in fact, unlike $S_\Gamma^{\rm good}$, it avoids introducing spurious poles on the physical sheet at singular breather kinematics **(c)**/**(d)**.

Table 2 summarises the classification of all five candidate families across both equations.

Table 2: Classification of two-soliton S-matrix candidate families against constraints (i)–(v) **(c)**/**(d)** (`scripts/smx_dictionary_and_families.py`, `scripts/smx_poles_good_vs_bad.py`, `scripts/smx_twisted_ring_bethe.py`).

Candidate family	(i)	(ii)	(iii)	(iv)	(v)	Status and verdict
I. Gamma $S_\Gamma^{(s=1)}$ (12)–(13)	Pass	Pass	Pass	Pass (bad) Fail* (good)	Pass	**(d)** baseline for bad eq; *spurious physical-sheet poles in good eq (Prop. 3.1)
Ia. Shifted Gamma $S_\Gamma^{(s=1/2)}$	Pass	Pass	Pass	Half-int.	Pass	Zero Maslov offset; shifts poles to $I_{\rm br} = (N - \frac{1}{2})\hbar$ **(c)**
Ib. η-CDD $S_\Gamma \cdot F_{\rm CDD}(\eta)$ (14)	Pass	Pass	Pass	Extra poles	Pass	Constant $O(\hbar^0)$ step as $\hbar \to 0$; adds/cancels poles in η **(c)**
II. $\hbar$-renormalised $S_\Gamma(\hbar_{\rm ren})$	Pass	Pass	Pass	$O(\hbar^2)$ shift	Pass	Both global and dynamical-only preserve (ii)–(iii); shifts $O(\hbar^0)$ by $-c_1\delta$ **(c)**
IIIa. Quantum dilog $\Phi_\hbar(f(D))\Phi_\hbar(g(S))$	Fail	Pass	—	—	—	Excluded at $O(\hbar^{-1})$ by $\partial_S\partial_D\delta_{\rm dyn} \neq 0$ **(c)**
IIIb. Mixed-arg. quantum dilog $\Phi_\hbar(A_{12})$, $\Phi_\hbar(R_{rs})$	Fail	—	—	—	—	Excluded at $O(\hbar^{-1})$ by weight-0 theorem and diagonal residue (Thm. 2.3) **(c)**
IV. Mirror product $F(D)G(S)e^{iT/\hbar}$	Fail	Fail	—	—	—	Excluded at $O(\hbar^{-1})$ by (i) and at *all* orders by (ii): $G(S) = \text{const}$ (Prop. 2.5) **(c)**
V. Dressed exponential $\exp(i\delta/\hbar + i\delta_1)$	Pass	Pass	Pass	Fail (bad) Pass (good)	Pass	Excluded for bad eq (no BC poles); viable **(d)** for good eq (no breathers)

3 Criterion (iv): bound-state poles in the bad vs. good equations

A decisive structural difference between the bad and good Boussinesq equations lies in their complex-conjugate two-soliton (breather) reductions.

3.1 Bad equation: Bogdan–Charkina breathers and physical-sheet poles

For the bad equation, take two right-movers ($\epsilon_1 = \epsilon_2 = +1$) at complex-conjugate rapidities $\beta_1 = a - ib$, $\beta_2 = a + ib$ with $a > 0$ and $b \in (\pi/3, \pi/2)$ (or the parity-image window $b \in (\pi/2, 2\pi/3)$). Then **(a)** [1] / **(c)** (`smatrix_poles_bootstrap.py`, `scripts/smx_poles_good_vs_bad.py`):

1. The interaction constant is real and exceeds unity precisely in this window:

$$A_{12}^{\rm bad}(a - ib, a + ib) = \frac{1 + 3\coth^2 a}{1 - 3\cot^2 b} > 1 \quad \Longleftrightarrow \quad \cot^2 b < \frac{1}{3} \quad \Longleftrightarrow \quad b \in \left(\frac{\pi}{3}, \frac{2\pi}{3}\right) \setminus \left\{\frac{\pi}{2}\right\}, \tag{24}$$

giving a regular, spatially localised periodic breather of envelope speed $V_{\rm br} = \cosh a \cos 2b / \cos b$ and total energy $E_{\rm br} = E(\beta_1) + E(\beta_2) \in \mathbb{R}$.

2. For $b \in (\pi/3, \pi/2)$, the relative momentum has positive imaginary part,

$$\mathrm{Im}(P_1 - P_2) = -\frac{1}{3}\sin 2b\left[\cosh 4a\cos 2b - \cosh 2a\right] > 0, \tag{25}$$

which defines the physical sheet for two-body bound states of $S(1,2)$, while

$$i\eta = \frac{2i}{9\hbar}(\ell_1 - \ell_2) = \frac{4}{9\hbar}\sinh 3a\sin 3b = -\frac{I_{\mathrm{br}}}{\hbar} < 0, \qquad I_{\mathrm{br}} = -\frac{4}{9}\sinh 3a\sin 3b > 0. \tag{26}$$

3. Consequently, the $\Gamma(1+i\eta)$ numerator of $S^{\mathrm{bad}}_{\Gamma,\mathrm{ov}}$ has simple poles on the physical sheet precisely at the Bogdan–Charkina Bohr–Sommerfeld quantisation levels $I_{\mathrm{br}} = N\hbar$ ($N = 1, 2, \dots$), and has no pole at the threshold $N = 0$ ($b \to \pi/3^+$) because $\Gamma(1) = 1$ is finite **(c)**.

3.2 Good equation: absence of regular breathers, spurious poles of $S^{\mathrm{good}}_{\Gamma}$, and Y-junction resonances

For the good equation, the situation is inverted:

Proposition 3.1 (**(a)** [8, 2] / **(c)**, `scripts/smx_poles_good_vs_bad.py`). *1.* ***Absence of regular Hirota breathers.*** *For any complex-conjugate pair $\alpha_1 = a - ib$, $\alpha_2 = a + ib$ with $a \in (0, \pi/2) \setminus \{\pi/6\}$ and $b \in \mathbb{R} \setminus \{0\}$ ($\epsilon_1 = \epsilon_2 = +1$), the Hirota interaction constant is*

$$A_{12}^{\mathrm{good}}(a - ib, a + ib) = \frac{1 - 3\cot^2 a}{1 + 3\coth^2 b} < \frac{1}{4}. \tag{27}$$

Specifically, $A_{12}^{\mathrm{good}} < 0$ for $a \in (0, \pi/6)$ and $0 < A_{12}^{\mathrm{good}} < 1/4$ for $a \in (\pi/6, \pi/2)$ (since $\coth^2 b > 1$).As shown in Proposition 3.2 of `review.pdf`*, the two-soliton τ-function $\tau_{\mathrm{br}} = 2e^{\theta_R}(\cos\theta_I + \sqrt{A_{12}}\cosh\theta_R)$ is strictly positive for all (x,t) if and only if $A_{12} > 1/4$. Thus* every *complex-conjugate breather of the good equation develops logarithmic singularities in $u = 2(\ln\tau)_{xx}$ and has infinite energy; the good Boussinesq equation has no regular breathers.*

2. ***Spurious physical-sheet poles of the naive Gamma continuation $S^{\mathrm{good}}_{\Gamma}$.*** *At $\alpha_{1,2} = a \mp ib$, the relative momentum imaginary part and the Gamma argument are*

$$\mathrm{Im}(P_1 - P_2) = \frac{1}{3}\sinh 2b\left[\cos 4a\cosh 2b - \cos 2a\right], \qquad i\eta = \frac{2i}{9\hbar}(\ell_1 - \ell_2) = \frac{4}{9\hbar}\sin 3a\sinh 3b. \tag{28}$$

On the "light" branch $a \in (0, \pi/3)$, taking $b < 0$ (with $\cosh 2b < \cos 2a/\cos 4a$ when $\cos 4a > 0$, or for all $b < 0$ when $a \in [\pi/8, \pi/3)$) gives $\sinh 2b < 0$ and $\cos 4a\cosh 2b - \cos 2a < 0$, so

$$\mathrm{Im}(P_1 - P_2) > 0 \quad (\textit{physical sheet}) \qquad \textit{and} \qquad i\eta = \frac{4}{9\hbar}\sin 3a\sinh 3b < 0. \tag{29}$$

Hence $\Gamma(1+i\eta)$ in $S^{\mathrm{good}}_{\Gamma,\mathrm{ov}}$ has simple poles on the physical sheet $\mathrm{Im}(P_1 - P_2) > 0$ at

$$\sinh 3b = -\frac{9N\hbar}{4\sin 3a} \qquad (N = 1, 2, \dots), \tag{30}$$

even though at every such root $A_{12}^{\mathrm{good}}(a - ib, a + ib) < 1/4$ corresponds to a singular *solution (for instance, at $\hbar = 0.05$ and $N = 1$, $a = 0.4$ gives $b = -0.04021$, $\mathrm{Im}(P_1 - P_2) = 0.01919 > 0$, and $A_{12}^{\mathrm{good}} = -2.177 < 0$; $a = 0.65$ gives $b = -0.04031$, $\mathrm{Im}(P_1 - P_2) = 0.05586 > 0$, and $A_{12}^{\mathrm{good}} = +0.02585 < 1/4$).*

3. ***Smoothness across real Y-junction resonances.*** *On the real rapidity square $(\alpha_1, \alpha_2) \in (0, \pi/2)^2$, the good equation has resonant Y-junction boundaries where $A_{12} = 0$ or $A_{12} = \infty$* **(a)** *[8, 2, 9]:*

- *Head-on ($\epsilon_1 = +1, \epsilon_2 = -1$): $\alpha_1 + \alpha_2 = \pi/3$ ($A_{12}^{\rm ho} = \infty$);*
- *Overtaking ($\epsilon_1 = \epsilon_2 = +1$): $|\alpha_1 - \alpha_2| = \pi/3$ ($A_{12}^{\rm ov} = \infty$) and $\alpha_1 + \alpha_2 = 2\pi/3$ ($A_{12}^{\rm ov} = 0$).*

On all three *real resonance lines, $\ell_1 - \ell_2 = 0$ ($\eta = 0$) identically: for head-on at $\alpha_1 + \alpha_2 = \pi/3$, $\ell_1 - \ell_2 = -\cos 3\alpha_1 - \cos(\pi - 3\alpha_1) = 0$; for overtaking at $\alpha_2 = \alpha_1 \pm \pi/3$ or $\alpha_2 = 2\pi/3 - \alpha_1$, $\ell_1 - \ell_2 = -\cos 3\alpha_1 + \cos 3\alpha_2 = 0$. Because $\lim_{\eta\to 0}(e^2 A_{12}/\eta^2)^{i\eta} = 1$ and $\Gamma(1) = 1$, $S_\Gamma^{\rm good}$ approaches the smooth topological phase $e^{iT_{12}/\hbar}$ on the entire Y-junction locus, with neither poles nor zeros there.*

Remark 3.2 (**(c)**/**(d)**). Proposition 3.1(2) shows that for the good equation, the naive analytic continuation $S_\Gamma^{\rm good}$ violates criterion (iv) unless either (a) the quantum good equation admits bound states without regular classical Hirota breather counterparts (a guess **(d)** with no known evidence), or (b) the good-equation S-matrix has its Gamma poles cancelled (as in Family V, $S_{\rm exp}^{\rm good}$) or moved off the physical sheet. By contrast, for the bad equation, the physical-sheet poles of $S_\Gamma^{\rm bad}$ match the regular Bogdan–Charkina breather spectrum without discrepancy.

4 Criterion (v): twisted N-soliton Bethe equations on a ring

Section 9 of `report.pdf` listed the asymptotic Bethe equations on a ring of circumference Λ—and specifically the ϕ-winding twist induced by the topological cross terms $\sum_{X_i > X_j} d\Pi_i \wedge dM_j$ in the N-soliton symplectic form—as not yet worked out. Here we derive the twisted N-soliton ring Bethe equations from the chamber-dependent symplectic geometry and verify their consistency properties at 40-digit precision.

4.1 N-soliton symplectic form and Darboux coordinates across ordering chambers

On the infinite line, an N-soliton state with asymptotic positions $X_1, \ldots, X_N$ lies in one of $N!$ spatial ordering chambers $\sigma \in S_N$, defined by $X_{\sigma(1)} < X_{\sigma(2)} < \cdots < X_{\sigma(N)}$. In chamber σ, the canonical symplectic form $\Omega = \int d\pi \wedge d\phi$ reduces on the asymptotic N-soliton manifold (with left-boundary reference $c = \phi(-\infty)$ and $\Pi_{\rm tot} = \sum_{j=1}^N \Pi_j$) to **(c)** (`report.pdf`, Prop. 3.1):

$$\Omega^{(\sigma)} = d\Pi_{\rm tot} \wedge dc + \sum_{j=1}^{N} dP_j \wedge dX_j + \sum_{\sigma^{-1}(i) > \sigma^{-1}(j)} d\Pi_i \wedge dM_j. \tag{31}$$

Define in chamber σ the Darboux position coordinates

$$X_j'^{(\sigma)} = X_j + \gamma_j \sum_{\sigma^{-1}(l) < \sigma^{-1}(j)} M_l, \qquad \gamma_j = \frac{d\Pi_j}{dP_j}. \tag{32}$$

Proposition 4.1 (**(c)**, `scripts/smx_twisted_ring_bethe.py`)**.** *For every $N \geq 2$ and every permutation $\sigma \in S_N$, in both the good and bad equations:*

1. *The canonical wedge sum $\sum_{j=1}^N dP_j \wedge dX_j'^{(\sigma)}$ equals $\sum_{j=1}^N dP_j \wedge dX_j + \sum_{\sigma^{-1}(i) > \sigma^{-1}(j)} d\Pi_i \wedge dM_j$, so $(c, \Pi_{\rm tot}; X_1'^{(\sigma)}, P_1, \ldots, X_N'^{(\sigma)}, P_N)$ are exact Darboux coordinates in chamber σ:*

$$\Omega^{(\sigma)} = d\Pi_{\rm tot} \wedge dc + \sum_{j=1}^{N} dP_j \wedge dX_j'^{(\sigma)}. \tag{33}$$

2. *Across any wall between adjacent chambers σ and $\sigma' = (j\ l)\sigma$ where solitons j (left) and l (right) cross, the Hirota scattering map $X_j^+ = X_j^- + \Delta x_{jl}$, $X_l^+ = X_l^- + \Delta x_{lj}$ ($\Delta x_{jl} =$*

$-\frac{1}{k_j}\ln A_{jl}$, $\Delta x_{lj} = +\frac{1}{k_l}\ln A_{jl}$) *shifts the Darboux coordinates by*

$$X_j'^{(\sigma')} - X_j'^{(\sigma)} = \Delta x_{jl} - \gamma_j M_l = \Delta x'_{jl} = -\frac{\partial\delta(j,l)}{\partial P_j}, \tag{34}$$
$$X_l'^{(\sigma')} - X_l'^{(\sigma)} = \Delta x_{lj} + \gamma_l M_j = \Delta x'_{lj} = -\frac{\partial\delta(j,l)}{\partial P_l},$$

and preserves the symplectic form, $\Omega^{(\sigma')} = \Omega^{(\sigma)}$*, by virtue of the identity*

$$P'_j\frac{\partial\Delta x_{jl}}{\partial\beta_l} - P'_l\frac{\partial\Delta x_{lj}}{\partial\beta_j} = -\Pi'_l M'_j - \Pi'_j M'_l. \tag{35}$$

Verified to 40 digits for $N = 4$ *across all* $4! = 24$ *ordering chambers and all* $\binom{4}{2} = 6$ *pairs in both the bad and good equations (maximum residual* $< 10^{-38}$*).*

4.2 Seam jump on the ring S^1_Λ and the twisted directional Bethe phase

Now place N well-separated solitons on a ring $x \in [0,\Lambda)$ of circumference $\Lambda \gg \max_j k_j^{-1}$, cut open at a reference seam $x = 0$ where $c = \phi(0^+)$. Each soliton j is a kink of the potential ϕ carrying topological charge $M_j = 2k_j$ and conjugate shift momentum $\Pi_j = -M_j v_j$, so the total winding across the ring is $|\phi(\Lambda^-) - \phi(0^+)| = \sum_{j=1}^N M_j = M_{\rm tot}$.

Follow soliton j as it traverses the ring once in the positive direction from $X_j = 0^+$ (where j is the leftmost soliton, $\sigma^{-1}(j) = 1$) to $X_j = \Lambda^-$ (where j is the rightmost soliton, $\sigma^{-1}(j) = N$), and then jumps back across the seam $\Lambda^- \to 0^+$:

1. **Bulk collisions ($0^+ \to \Lambda^-$):** In the interior of the interval $(0,\Lambda)$, soliton j overtakes or crosses every other soliton $l \neq j$. In the canonical Darboux coordinates $X'^{(\sigma)}$, each crossing of j past l is generated by the two-body phase $\delta(j,l)$ (or $\hbar\arg S(j,l)$ in the quantum theory), which encodes the Darboux shift $\Delta x'_{jl} = -\partial_{P_j}\delta(j,l)$.

2. **Seam transition ($\Lambda^- \to 0^+$):** When soliton j passes from $X_j = \Lambda^-$ across the seam to $X_j = 0^+$, two topological jumps occur simultaneously:

   - First, all $N-1$ other solitons $l \neq j$, which lay to the *left* of j at $X_j = \Lambda^-$, now lie to the *right* of j at $X_j = 0^+$. By (32), the Darboux coordinate X'_j jumps across the seam by
   $$\Delta_{\rm seam}X'_j = X'_j\big|_{X_j=\Lambda^-} - X'_j\big|_{X_j=0^+} = \Lambda + \gamma_j\sum_{l\neq j} M_l. \tag{36}$$
   Integrating $P_j\, d(\Delta_{\rm seam}X'_j)$ at fixed M_l gives the kinetic-plus-Darboux seam action $P_j\Lambda + \Pi_j\sum_{l\neq j}M_l$.
   - Second, passing the kink j across the seam $x = 0$ shifts the seam potential $c = \phi(0^+)$ by M_j, which couples to the conjugate momentum Π_j to contribute the self-winding phase $\Pi_j M_j$.

Combining the bulk scattering phase $\delta(j,l)$ with the pairwise topological seam twist $+\Pi_j M_l$ defines the *twisted directional ring Bethe phase* of soliton j passing soliton l:

$$\Phi_{\rm ring}(j\mid l) \coloneqq \delta(j,l) + \Pi_j M_l = \frac{2}{9}(\ell_j - \ell_l)\ln A_{jl} + \frac{1}{3}\Pi_j M_l + \frac{2}{3}\Pi_l M_j. \tag{37}$$

For any candidate quantum S-matrix $S(j,l;\hbar)$, the quantum twisted ring Bethe equations are therefore

$$\mathcal{B}_j^{\rm ring} \coloneqq P_j\Lambda + \Pi_j M_j + \sum_{l\neq j}\Big(\hbar\arg S(j,l;\hbar) + \Pi_j M_l\Big) = 2\pi\hbar n_j, \qquad n_j \in \mathbb{Z} \quad (j = 1,\dots,N), \tag{38}$$

or in multiplicative form,

$$\exp\Big(\frac{i}{\hbar}\big(P_j\Lambda + \Pi_j M_{\rm tot}\big)\Big)\prod_{l\neq j} S(j,l;\hbar) = 1 \qquad (j=1,\dots,N). \tag{39}$$

Theorem 4.2 (**(c)**, `scripts/smx_twisted_ring_bethe.py`)**.** *The twisted directional ring Bethe phase* (37) *and the ring Bethe equations* (38)–(39) *satisfy three exact consistency laws across both the bad and good equations and in both overtaking and head-on channels:*

1. ***Exact recovery of the physical Hirota shift*** Δx_{jl}**.** *Differentiating* $\Phi_{\rm ring}(j \mid l)$ *with respect to the physical momentum* P_j *at fixed soliton* l *gives*

$$-\frac{\partial\Phi_{\rm ring}(j\mid l)}{\partial P_j} = -\frac{\partial\delta(j,l)}{\partial P_j} - \gamma_j M_l = \Delta x'_{jl} + \gamma_j M_l = \Delta x_{jl} = -\frac{1}{k_j}\ln A_{jl}. \tag{40}$$

Thus the classical trajectory shift in the ring Bethe equation $\partial_{P_j}\mathcal{B}_j^{\rm ring} = \Lambda - \sum_{l\neq j}\Delta x_{jl}$ *is the* exact physical Hirota shift Δx_{jl}*. By contrast:*

- *Omitting the seam twist* $+\Pi_j M_l$ *gives* $-\partial_{P_j}\delta(j,l) = \Delta x_{jl} - \gamma_j M_l$*, which misses* Δx_{jl} *by the* $O(1)$ *Darboux shift* $-\gamma_j M_l$ *(e.g. error* 0.680 *at* $(\beta_1,\beta_2) = (1.1, 0.4)$*);*
- *Omitting the topological term* T_{jl} *altogether gives* $-\partial_{P_j}\delta_{\rm dyn}(j,l) = \Delta x_{jl} + (\frac{1}{3}\Pi'_j M_l + \frac{2}{3}M'_j\Pi_l)/P'_j$*, which also fails by* $O(1)$ *(e.g. error* 0.916 *at* $(\beta_1,\beta_2) = (1.1, 0.4)$*).*

Verified to 40 digits at six test pairs across both equations and channels (residual $< 10^{-38}$*).*

2. ***Global translation-winding sum rule.*** *For any unitary candidate* $S(j,l;\hbar)S(l,j;\hbar) = 1$*,* $\arg S(j,l;\hbar) + \arg S(l,j;\hbar) = 0$*, so the pairwise twisted phases satisfy the non-antisymmetric reciprocity relation*

$$\Phi_{\rm ring}(j\mid l) + \Phi_{\rm ring}(l\mid j) = \Pi_j M_l + \Pi_l M_j \neq 0. \tag{41}$$

Summing the twisted Bethe equations (38) *over* $j = 1,\dots,N$ *causes all bulk S-matrix phases* $\arg S(j,l;\hbar)$ *to cancel pairwise while the diagonal and off-diagonal topological twists combine into the total bilinear* $\Pi_{\rm tot}M_{\rm tot}$*:*

$$\sum_{j=1}^{N}\mathcal{B}_j^{\rm ring} = P_{\rm tot}\Lambda + \Pi_{\rm tot}M_{\rm tot} = 2\pi\hbar\sum_{j=1}^{N} n_j, \quad P_{\rm tot} = \sum_{j=1}^{N}P_j, \;\; \Pi_{\rm tot} = \sum_{j=1}^{N}\Pi_j, \;\; M_{\rm tot} = \sum_{j=1}^{N}M_j. \tag{42}$$

This is the exact field-theoretic momentum quantization law for a state of total field momentum $P_{\rm tot}$ *and zero-mode momentum* $\Pi_{\rm tot}$ *in the winding sector* $\Delta\phi = M_{\rm tot}$ *on* S^1_Λ*.*

3. ***Covariant fusion onto Bogdan–Charkina breathers.*** *For a bad-equation breather with constituent rapidities* $\beta_{1,2} = a \mp ib$ *(*$\epsilon_1 = \epsilon_2 = +1$*) and a third soliton* β_3 *of either direction* $\epsilon_3 = \pm 1$*, the constituent masses and shift charges sum to real composite breather charges:*

$$M_{\rm br} = M_1 + M_2 = 4\sinh a\cos b \in \mathbb{R}, \qquad \Pi_{\rm br} = \Pi_1 + \Pi_2 = -2\sinh 2a\cos 2b = -M_{\rm br}V_{\rm br} \in \mathbb{R}, \tag{43}$$

where $V_{\rm br} = \cosh a\cos 2b/\cos b$ *is the classical breather envelope velocity. Consequently, the fused ring Bethe phase* $\Phi_{\rm ring}(br\mid 3) = \Phi_{\rm ring}(1\mid 3) + \Phi_{\rm ring}(2\mid 3) = \delta(1,3) + \delta(2,3) + \Pi_{\rm br}M_3$ *is real to 40 digits, and*

$$-\frac{\partial}{\partial P_3}\big[\Phi_{\rm ring}(3\mid 1) + \Phi_{\rm ring}(3\mid 2)\big] = -\frac{1}{k_3}\ln(A_{31}A_{32}) = \Delta x_{3,\rm br} \tag{44}$$

reproduces the exact classical breather–soliton Hirota shift (verified to 40 digits for both $\epsilon_3 = +1$ *and* $\epsilon_3 = -1$*).*

5 Minimal residual ambiguity and precise interface to other packages

5.1 Minimal residual ambiguity after constraints (i)–(v)

We now determine the exact freedom that remains after imposing (i) the semiclassical limit, (ii) unitarity, (iii) quantum crossing, (iv) the Bogdan–Charkina integer breather spectrum $I_{\rm br} = N\hbar$ ($N \geq 1$, no $N = 0$ pole) for the bad equation, and (v) consistency with the twisted ring Bethe equations (38).

Proposition 5.1 (**(c)**, `scripts/smx_residual_ambiguity_interface.py`)**.** *For the bad equation, let* $S_{\Gamma,\rm ov} = S_{\rm ov}^{(s=1)}$ *and* $S_{\Gamma,\rm ho} = S_{\rm ho}^{(s=1)}$ *be the baseline Gamma candidate* (12)–(13)*. Any candidate two-soliton S-matrix* $S(1,2;\hbar)$ *that satisfies (i)–(v) with simple physical-sheet poles solely at* $I_{\rm br} = N\hbar$ *(*$N \geq 1$*) takes the form*

$$S_{\rm ov}(1,2;\hbar) = S_{\Gamma,\rm ov}(1,2;\hbar)\,\exp\big(i\Theta_{\rm ov}(1,2;\hbar)\big), \qquad S_{\rm ho}(1,2;\hbar) = S_{\Gamma,\rm ho}(1,2;\hbar)\,\exp\big(i\Theta_{\rm ho}(1,2;\hbar)\big), \tag{45}$$

where $\Theta_{\rm ov}, \Theta_{\rm ho}$ *have asymptotic expansions* $\Theta_{\rm ov/ho}(1,2;\hbar) = \vartheta_{\rm ov/ho}(1,2) + \hbar\,\vartheta^{(1)}_{\rm ov/ho}(1,2) + O(\hbar^2)$ *as* $\hbar \to 0$ *at fixed rapidities, are holomorphic in the physical breather strip* $0 < \mathrm{Im}(-i\eta) < 1$*, and obey:*

1. ***Real-valuedness and antisymmetry (from unitarity (ii)):*** *For real rapidities,* $\Theta_{\rm ov}, \Theta_{\rm ho} \in \mathbb{R}$ *and*

$$\Theta_{\rm ov}(1,2;\hbar) + \Theta_{\rm ov}(2,1;\hbar) = 0, \qquad \Theta_{\rm ho}(1,2;\hbar) + \Theta_{\rm ho}(2,1;\hbar) = 0 \pmod{2\pi}. \tag{46}$$

2. ***Crossing locking of the head-on channel (from crossing (iii)):*** *Because* $S_{\Gamma,\rm ov}(\beta_1,\beta_2 + i\pi;\hbar)S_{\Gamma,\rm ho}(\beta_1,\beta_2;\hbar) = 1$ *holds identically,* $\Theta_{\rm ho}$ *is completely fixed by analytic continuation of* $\Theta_{\rm ov}$*:*

$$\Theta_{\rm ho}(\beta_1,\beta_2;\hbar) = -\Theta_{\rm ov}(\beta_1,\beta_2 + i\pi;\hbar) \pmod{2\pi}. \tag{47}$$

3. ***Automatic ring Bethe compatibility (criterion (v)):*** *Any* $(\Theta_{\rm ov}, \Theta_{\rm ho})$ *satisfying* (46) *preserves the global ring Bethe sum rule* (42) *identically at all orders in* $\hbar$*, and shifts the quantum ring Bethe phase at* $O(\hbar)$ *without altering the classical Hirota shift* (40)*.*

Consequently, at one-loop order $O(\hbar^0)$*, the* minimal residual ambiguity *after (i)–(v) is a* ***single real antisymmetric two-variable function*** $\vartheta_{\rm ov}(\beta_1,\beta_2) = -\vartheta_{\rm ov}(\beta_2,\beta_1)$ *in the overtaking channel (with* $\vartheta_{\rm ho}(\beta_1,\beta_2) = -\vartheta_{\rm ov}(\beta_1,\beta_2 + i\pi)$ *locked by crossing).*

5.2 Interface to the projected determinant (`det_*`) and quantum frame (`frame_*`) packages

Write the full $O(\hbar^0)$ one-loop phase of the physical S-matrix as

$$\delta_1^{\rm phys}(1,2) := \lim_{\hbar\to 0}\Big(\arg S(1,2;\hbar) - \frac{\delta(1,2)}{\hbar}\Big) = \pm\frac{\pi}{2}\,\mathrm{sgn}(\eta) + \vartheta_{\rm ov/ho}(1,2) = \delta_1^{\rm det}(1,2) + \delta_1^{\rm frame}(1,2), \tag{48}$$

where $\delta_1^{\rm det}(1,2)$ is the intrinsic one-loop phase from the projected fluctuation determinant (`det_*` package) and $\delta_1^{\rm frame}(1,2)$ is the conversion from bare classical rapidities to renormalised quantum physical momenta (`frame_*` package). We now establish the exact crossing and small-amplitude laws governing both pieces.

1. Continuum one-loop integrals and the cusp-jump cancellation of the Stirling offset. In `oneloop_phase.pdf` (and under conditional status in `audit_determinant_foundations.pdf`), the raw continuum integral over phonon phase shifts on the good-equation two-soliton background is

$$\Delta\theta_{\rm cont}^{\rm same/head}(\alpha_1,\alpha_2) = I_{12}^{\rm same/head}(\alpha_1,\alpha_2) + J_{12}^{\rm same/head}(\alpha_1,\alpha_2), \tag{49}$$

where, with $\varphi_{j,r} \in \{\alpha_j + 2\pi/3, \alpha_j - 2\pi/3, \alpha_j\}$ and $\sigma_r \in \{+1,+1,-1\}$:

$$I_{12}^{\rm same}(\alpha_1,\alpha_2) = \frac{1}{\pi}\sum_{r,s=1}^{3}\sigma_r\sigma_s\Big[\mathrm{Li}_2(-R_{rs}) - \mathrm{Li}_2(R_{rs}) + \ln|R_{rs}|\ln\frac{|1-R_{rs}|}{1+|R_{rs}|}\Big], \quad R_{rs} = \frac{\tan(\varphi_{1,r}/2)}{\tan(\varphi_{2,s}/2)}, \tag{50}$$

$$J_{12}^{\rm same}(\alpha_1,\alpha_2) = \sum_{s=1}^{3}\sigma_s\,\mathrm{sgn}(\cos\varphi_{2,s})\arctan\big(\tan\alpha_1\cot\varphi_{2,s}\big) - (\alpha_1 \leftrightarrow \alpha_2), \tag{51}$$

and in the head-on channel R_{rs} is replaced by $\tilde{R}_{rs} = \tan(\varphi_{1,r}/2)\tan(\varphi_{2,s}/2)$ and $\cot\varphi_{2,s}$ by $-\tan\varphi_{2,s}$ in the $1\to 2$ term (`oneloop_phase.pdf`, Eqs. (16)–(19)).

Theorem 5.2 (**(c)**, `scripts/smx_residual_ambiguity_interface.py`)**.** *For all $(\alpha_1,\alpha_2) \in (0,\pi/3)^2$ (the sub-radiation-threshold square where $\cos 3\alpha_j > 0$), the continuum integrals (50)–(51) satisfy the exact crossing identities*

$$I_{12}^{\rm same}(\alpha_1,\alpha_2+\pi) = -I_{12}^{\rm head}(\alpha_1,\alpha_2), \qquad J_{12}^{\rm same}(\alpha_1,\alpha_2+\pi) + J_{12}^{\rm head}(\alpha_1,\alpha_2) = +\frac{\pi}{2}, \tag{52}$$

and hence

$$\Delta\theta_{\rm cont}^{\rm same}(\alpha_1,\alpha_2+\pi) + \Delta\theta_{\rm cont}^{\rm head}(\alpha_1,\alpha_2) = +\frac{\pi}{2}. \tag{53}$$

Verified to 40 digits across $(0,\pi/3)^2$ (maximum residual $< 10^{-39}$).

Proof. Under $\alpha_2 \mapsto \alpha_2+\pi$, each angle shifts by $\varphi_{2,s} \mapsto \varphi_{2,s}+\pi$, so $\tan(\varphi_{2,s}/2) \mapsto -\cot(\varphi_{2,s}/2)$, $R_{rs} \mapsto -\tilde{R}_{rs}$, and $I_{12}^{\rm same}(\alpha_1,\alpha_2+\pi) = -I_{12}^{\rm head}(\alpha_1,\alpha_2)$ term by term. In J_{12}, let $f_1(\alpha_2) = \sum_s \sigma_s\,\mathrm{sgn}(\cos\varphi_{2,s})\arctan(\tan\alpha_1\cot\varphi_{2,s})$ be the smooth function on $(0,\pi/3)$ (where $\cos\varphi_{2,1} < 0, \cos\varphi_{2,2} < 0, \cos\varphi_{2,3} > 0$). Because $\cot(\varphi_{2,s}+\pi) = \cot\varphi_{2,s}$ is π-periodic whereas $\mathrm{sgn}(\cos(\varphi_{2,s}+\pi)) = -\mathrm{sgn}(\cos\varphi_{2,s})$, continuing f_1 analytically from $(0,\pi/3)$ across the radiation-threshold cusp $\alpha_2 = \pi/3$ (where $J_{12}^{\rm same}$ jumps by $+\pi$) to $\alpha_2+\pi \in (\pi,4\pi/3)$ reverses the sign of f_1. Meanwhile, in $J_{12}^{\rm head}$, the identity $\arctan(x) - \arctan(1/x) = \frac{\pi}{2}\mathrm{sgn}(x)$ applied to each of the three s-terms with signs $(\sigma_1,\sigma_2,\sigma_3)\cdot(-1,-1,+1) = (-1,-1,-1)$ shifts the sum of the three $\pm\pi/2$ constants by $(-1-1+1)\frac{\pi}{2} = -\frac{\pi}{2}$, yielding $J_{12}^{\rm same}(\alpha_1,\alpha_2+\pi) + J_{12}^{\rm head}(\alpha_1,\alpha_2) = +\frac{\pi}{2}$. □

Corollary 5.3 (Cancellation of the head-on Stirling offset under crossing, **(c)**)**.** *For $\alpha_1 < \alpha_2 \in (0,\pi/3)$ ($\ell_1 < \ell_2$, $\eta < 0$), the Gamma candidate $S_\Gamma^{(s=1)}$ has Stirling $O(\hbar^0)$ offset $-\frac{\pi}{2}$ in overtaking and $+\frac{\pi}{2}$ in head-on ($\eta_{\rm ho} > 0$). By Theorem 5.2, an overtaking one-loop phase of the form*

$$\delta_{1,\rm ov}^{\rm det}(\alpha_1,\alpha_2) = -c_{\rm M}\frac{\pi}{2} + c_{\rm loop}\Delta\theta_{\rm cont}^{\rm same}(\alpha_1,\alpha_2) \qquad (0<\alpha_1<\alpha_2<\pi/3) \tag{54}$$

continues under exact crossing $\delta_{1,\rm ho}^{\rm det}(\alpha_1,\alpha_2) = -\delta_{1,\rm ov}^{\rm det}(\alpha_1,\alpha_2+\pi)$ to

$$\delta_{1,\rm ho}^{\rm det}(\alpha_1,\alpha_2) = \Big(c_{\rm M} - \frac{1}{2}c_{\rm loop}\Big)\frac{\pi}{2} + c_{\rm loop}\Delta\theta_{\rm cont}^{\rm head}(\alpha_1,\alpha_2). \tag{55}$$

In particular, for $c_{\rm loop} = 1$:

- *If $c_{\rm M} = 1$ (matching the overtaking Stirling offset $-\frac{\pi}{2}$ of $S_{\Gamma,\rm ov}$), the crossed head-on threshold constant is $(1-\frac{1}{2})\frac{\pi}{2} = +\frac{\pi}{4}$ for $\Delta\theta_{\rm cont}$ or, if only I_{12} is crossed and J_{12} is evaluated on its principal branch (jumping by $+\pi$ at $\alpha_2 = \pi/3$), $(c_{\rm M}-1)\frac{\pi}{2} = 0$ in the head-on channel!*

Thus the $+\frac{\pi}{2}$ crossing offset of J_{12} and the $+\frac{\pi}{2}$ Stirling offset of $S_{\Gamma,\rm ho}$ are locked by the same threshold anomaly.

2. Frame-conversion interface and small-amplitude scaling law. Suppose the quantum momentum/frame dictionary (`frame_*`) supplies a single-soliton one-loop momentum shift $\Delta P(\alpha,\epsilon) = \epsilon\Delta P(\alpha,+1)$ (so $P_{\rm quant}(\alpha,\epsilon) = P(\alpha,\epsilon) + \hbar\Delta P(\alpha,\epsilon) + O(\hbar^2)$) and a one-loop coupling renormalisation c_1 ($\hbar_{\rm ren} = \hbar(1+c_1\hbar)$). Expanding $\delta(P_1^{\rm quant}, P_2^{\rm quant})/\hbar_{\rm ren}$ at fixed quantum momenta shifts the $O(\hbar^0)$ phase by

$$\delta_1^{\rm frame}(1,2) = \Big[\Delta x'_{12}\Delta P(\alpha_1,\epsilon_1)+\Delta x'_{21}\Delta P(\alpha_2,\epsilon_2)\Big]-c_1\delta(1,2) = \delta_1^{\rm Hirota}(1,2)+\delta_1^{\rm Darboux}(1,2)-c_1\delta(1,2), \tag{56}$$

where we split $\Delta x'_{12} = \Delta x_{12} - \gamma_1 M_2$ into the Hirota-shift piece and the Darboux-frame piece:

$$\begin{aligned}\delta_1^{\rm Hirota}(1,2) &= \Delta x_{12}\Delta P(\alpha_1,\epsilon_1) + \Delta x_{21}\Delta P(\alpha_2,\epsilon_2),\\ \delta_1^{\rm Darboux}(1,2) &= -\gamma_1 M_2\Delta P(\alpha_1,\epsilon_1) + \gamma_2 M_1\Delta P(\alpha_2,\epsilon_2).\end{aligned} \tag{57}$$

Proposition 5.4 (**(c)**, `scripts/smx_residual_ambiguity_interface.py`). *1. For* any *single-variable function $\Delta P(\alpha,+1)$ extended to crossed rapidities by $\Delta P(\alpha+\pi,+1) = -\Delta P(\alpha,-1) = \Delta P(\alpha,+1)$, both $\delta_1^{\rm Hirota}(1,2)$ and $\delta_1^{\rm Darboux}(1,2)$ are antisymmetric under $1\leftrightarrow 2$ and obey exact crossing $\delta^{\bullet}_{1,\rm ov}(\alpha_1,\alpha_2+\pi) = -\delta^{\bullet}_{1,\rm ho}(\alpha_1,\alpha_2)$ (verified to 40 digits).*

2. Assume $\Delta P(\alpha,+1)$ has the small-amplitude expansion

$$\Delta P(\alpha,+1) = c_{\ln}\,\alpha\ln\alpha + c_{\rm lin}\,\alpha + O(\alpha^3\ln\alpha) \qquad (\alpha\to 0^+). \tag{58}$$

Along any ray $\alpha_1 = \lambda r$, $\alpha_2 = \lambda$ with fixed ratio $r\in(0,1)$ as $\lambda\to 0^+$:

- *In the **overtaking channel** ($\epsilon_1=\epsilon_2=+1$), where $\Delta x_{12} = -\frac{1}{\lambda r}\ln\big(\frac{1-r}{1+r}\big)^2 + O(\lambda)$ and $-\gamma_1 M_2 = \frac{2}{\lambda r} + O(\lambda)$, both pieces have finite, scale-invariant $\lambda\to 0^+$ limits:*

$$\lim_{\lambda\to 0^+}\delta_{1,\rm ov}^{\rm Hirota}(\lambda r,\lambda) = -2c_{\ln}\ln r\ln\frac{1-r}{1+r}, \tag{59}$$

$$\lim_{\lambda\to 0^+}\delta_{1,\rm ov}^{\rm Darboux}(\lambda r,\lambda) = 2c_{\ln}\,r\ln r + 2c_{\rm lin}(r-1), \tag{60}$$

  while $\delta_{\rm ov}(\lambda r,\lambda) = O(\lambda^2\ln\lambda)\to 0$ (so c_1 drops out at $\lambda\to 0^+$) and $J_{12}^{\rm same}(\lambda r,\lambda) = O(\lambda^2)\to 0$. Meanwhile, the continuum dilogarithm integral (50) has the exact KdV limit

$$\lim_{\lambda\to 0^+} I_{12}^{\rm same}(\lambda r,\lambda) = -\frac{4}{\pi}\chi_2(r) - \frac{2}{\pi}\ln r\ln\frac{1-r}{1+r}, \qquad \chi_2(r) = \tfrac{1}{2}\big({\rm Li}_2(r) - {\rm Li}_2(-r)\big). \tag{61}$$

  Combining (59) and (61), the non-Legendre logarithmic product $\ln r\ln\frac{1-r}{1+r}$ in $c_{\rm loop}I_{12}^{\rm same}\pm\delta_{1,\rm ov}^{\rm Hirota}$ cancels if and only if $\pm c_{\ln} = c_{\rm loop}/\pi$ (for $c_{\rm loop}=1$, $c_{\ln} = \pm 1/\pi$, where the continuum mode integral $\Delta P_{\rm mode}(\alpha) = -\frac{2}{\pi}\alpha\ln\alpha + O(\alpha)$ has $c_{\ln} = -2/\pi$ and therefore cancels $\ln r\ln\frac{1-r}{1+r}$ in $I_{12}^{\rm same} - \delta_{1,\rm ov}^{\rm Hirota}[\Delta P_{\rm mode}] = -\frac{4}{\pi}\chi_2(r)$, independently confirming the sign audit in `matching.pdf`).

- *In the **head-on channel** ($\epsilon_1=+1,\epsilon_2=-1$), $A_{12}^{\rm ho} = 1 - \frac{3}{2}\lambda^2(1-r^2) + O(\lambda^4)$ gives $\Delta x_{12} = O(\lambda)$, so*

$$\delta_{1,\rm ho}^{\rm Hirota}(\lambda r,\lambda) = O(\lambda^2\ln\lambda)\longrightarrow 0, \qquad I_{12}^{\rm head}(\lambda r,\lambda) + J_{12}^{\rm head}(\lambda r,\lambda) = O(\lambda^2\ln\lambda)\longrightarrow 0, \tag{62}$$

  whereas $-\gamma_1 M_2 = \frac{2}{\lambda r} + O(\lambda)$ and $+\gamma_2 M_1 = \frac{2r}{\lambda} + O(\lambda)$ have the same *sign, so the Darboux-frame piece is*

$$\delta_{1,\rm ho}^{\rm Darboux}(\lambda r,\lambda) = 2c_{\ln}\big[(1+r)\ln\lambda + r\ln r\big] + 2c_{\rm lin}(1+r) + O(\lambda^2\ln\lambda), \tag{63}$$

  which vanishes as $\lambda\to 0^+$ if and only if $c_{\ln} = c_{\rm lin} = 0$ (or if the quantum frame conversion takes place directly in the Darboux momentum coordinates rather than via $\Delta x'\Delta P$).

Precise interface summary. Once the `det_*` and `frame_*` packages supply:

1. the intrinsic projected determinant phase $\delta^{\mathrm{det}}_{1,\mathrm{ov}}(\beta_1,\beta_2)$ (specifically the two dimensionless numbers $(c_{\mathrm{M}}, c_{\mathrm{loop}})$ in (54) and any zero-mode projection correction $\Delta\vartheta_{\mathrm{proj}}(\beta_1,\beta_2)$), and
2. the single-soliton one-loop momentum shift function $\Delta P(\beta,+1)$ (with small-amplitude coefficients $(c_{\ln}, c_{\mathrm{lin}})$ in (58)) and coupling shift c_1 in (56),

the residual $O(\hbar^0)$ function $\vartheta_{\mathrm{ov}}(1,2)$ in Proposition 5.1 is uniquely fixed with zero remaining free parameters by

$$\vartheta_{\mathrm{ov}}(1,2) = -\frac{\pi}{2}\,\mathrm{sgn}(\eta) + \delta^{\mathrm{det}}_{1,\mathrm{ov}}(1,2) + \Delta x'_{12}\Delta P(\beta_1,+1) + \Delta x'_{21}\Delta P(\beta_2,+1) - c_1\delta_{\mathrm{ov}}(1,2), \quad (64)$$

and $\vartheta_{\mathrm{ho}}(\beta_1,\beta_2) = -\vartheta_{\mathrm{ov}}(\beta_1,\beta_2+i\pi)$.

6 Summary of claims, verification scripts, and failures recorded

6.1 Summary table of claims and checks

Every verification script in `phase2/scripts/smx_*.py` runs in a few seconds at 35–40 digit precision (`mpmath`) and ends by printing `ALL CHECKS PASSED`.

6.2 Failures recorded

In accordance with `AGENTS.md`, we record all failed checks and intermediate errors encountered during this investigation, including those that were diagnosed and fixed:

1. **Swapped K^4 dispersion sign in the initial definition of A_{12} (`scripts/smx_dictionary_and_families.py`).** In the first draft of `A12(s1, s2)`, we wrote `W**2 - K**2 - sg * K**4` with `sg = -1` for the bad equation and `+1` for the good equation, which reversed the signs of K^4 between the two equations. This caused `scripts/smx_poles_good_vs_bad.py` to report $A_{12}^{\mathrm{good}} = 473.5 > 1/4$ at $(a,b) = (0.25, 0.2)$ and caused the symplectic and ring Bethe derivative checks in `scripts/smx_twisted_ring_bethe.py` and `scripts/smx_residual_ambiguity_interface.py` to fail. Fixing the polynomial to `W**2 - K**2 + sg * K**4` resolved all three failures immediately.

2. **Missing $-\frac{2}{3}M_1'\Pi_2$ term in the log-derivative identity (8).** We initially conjectured $\frac{2}{9}(\ell_1-\ell_2)\partial_1 \ln A_{12} = -\frac{1}{3}\Pi_1' M_2$, overlooking that M_1 also depends on β_1 inside $T_{12} = -\frac{2}{3}(\Pi_1 M_2 - \Pi_2 M_1)$. Numerical differentiation exposed the discrepancy; restoring the $-\frac{2}{3}M_1'\Pi_2$ term gave agreement to 10^{-38}. In `scripts/smx_twisted_ring_bethe.py`, at the head-on test pair $(\beta_1,\epsilon_1) = (0.9,+1)$, $(\beta_2,\epsilon_2) = (0.35,-1)$, the two terms $-\frac{1}{3}\Pi_1' M_2$ and $-\frac{2}{3}M_1'\Pi_2$ have opposite signs and nearly cancel, giving $|\partial_{P_1}\delta_{\mathrm{dyn}} + \Delta x_{12}| = 6.04\times 10^{-3}$ (smaller than our initial naive lower-bound threshold 10^{-2}, which was updated to the exact analytical prediction $|(\frac{1}{3}\Pi_1' M_2 + \frac{2}{3}M_1'\Pi_2)/P_1'|$).

3. **Sign of the $I_{\mathrm{KdV}}(r)$ vs. $\Delta x\,\Delta P_{\mathrm{mode}}$ log cancellation (`scripts/smx_residual_ambiguity_interface.py`).** We initially tested whether $I_{\mathrm{KdV}}(r) + \delta^{\mathrm{Hirota}}_{1,\mathrm{ov}}[\Delta P_{\mathrm{mode}}]$ (with a plus sign) equals $-\frac{4}{\pi}\chi_2(r)$ for $\Delta P_{\mathrm{mode}}(\alpha) = -\frac{2}{\pi}\alpha\ln\alpha$. That check failed with error 0.902 at $r = 0.25$ because both $I_{\mathrm{KdV}}(r)$ and $\delta^{\mathrm{Hirota}}_{1,\mathrm{ov}}[\Delta P_{\mathrm{mode}}]$ carry $-\frac{2}{\pi}\ln r \ln\frac{1-r}{1+r}$ with the *same* minus sign, so the logarithm cancels in the difference $I_{\mathrm{KdV}}(r) - \delta^{\mathrm{Hirota}}_{1,\mathrm{ov}}[\Delta P_{\mathrm{mode}}] = -\frac{4}{\pi}\chi_2(r)$ (independently confirming the sign correction recorded in `matching.pdf`).

4. **SymPy `simplify` hang on $\partial_{\beta_1}\partial_{\beta_2}\delta_{\mathrm{dyn}}$ (`scripts/smx_dictionary_and_families.py`).** Calling `sp.simplify(sp.diff(ddyn_sym, B1, B2))` directly on the hyperbolic expression timed out (> 3 minutes) because SymPy attempted hyperbolic trigonometric factorisation of high-degree polynomials. Expanding $\partial_1\partial_2[(\ell_1-\ell_2)\ln A_{12}] = \ell_1'\partial_2 \ln A_{12} - \ell_2'\partial_1 \ln A_{12} + (\ell_1-\ell_2)\partial_1\partial_2 \ln A_{12}$ directly verified the absence of `log` and `polylog` nodes in < 0.1 s.

5. **Binary float64 rounding in $\mathfrak{sl}_3$ test arguments and sign of the $O(\eta^{-1})$ Bernoulli term in Check 2 (`scripts/smx_dictionary_and_families.py`).** Passing Python `float` literals $(0.85 + 0.3)/2$ into `G_sl3_bad` introduced a 10^{-16} IEEE 754 rounding discrepancy against 40-digit `mpf` arithmetic, and the $O(\eta^{-1})$ Stirling term in (15) was initially written with $+B_2(s)/\eta$ instead of $-B_2(s)/\eta$ (since $\mathrm{Im}[B_2(s)/(2i\eta)] = -B_2(s)/(2\eta)$). Both were caught by the 30-digit assertion in `scripts/smx_dictionary_and_families.py` and corrected.

References

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Table 3: Summary of claims, labels, and verification scripts.

Claim / Formula	Label	Status	Verification script
Explicit good/bad dictionaries, KP cross-ratio (7), $\mathfrak{sl}_3$ factorisation $A_{\rm ov} = G(S/2)/G(D/2)$, and log-derivative identity (8)	**(c)**	Passed ($< 10^{-30}$)	`scripts/smx_dictionary_and_families.py`
Family I: Stirling expansion (15) for $s = 1, 1/2$ and constant $O(\hbar^0)$ step (16) of η-CDD factors	**(c)**	Passed ($< 10^{-11}$)	`scripts/smx_dictionary_and_families.py`
Family II: Separate crossing (17) of $\delta_{\rm dyn}$ and T_{12}; global and dynamical-only $\hbar$-renormalisation (18)	**(c)**	Passed ($< 10^{-30}$)	`scripts/smx_dictionary_and_families.py`
Family III: Weight-0 theorem for $\partial_1\partial_2\delta(1,2)$, exclusion of mixed-arg. dilogarithms $\mathrm{Li}_2(A_{12})$ and KP root-ratio basis $\mathrm{Li}_2(R_{rs})$ at $O(\hbar^{-1})$ (Thm. 2.3)	**(c)**	Passed (SymPy + lstsq)	`scripts/smx_dictionary_and_families.py`
Family IV: All-orders exclusion of mirror products $F(D)G(S)$ by exchange unitarity $G(S) = \text{const}$ (Prop. 2.5)	**(c)**	Passed (SymPy + 40d)	`scripts/smx_dictionary_and_families.py`
Bad eq. physical-sheet breather poles at $I_{\rm br} = N\hbar$ ($N \geq 1$) and regularity at $N = 0$	**(c)**	Passed ($< 10^{-32}$)	`scripts/smx_poles_good_vs_bad.py`
Good eq.: $A_{12}^{\rm good}(a - ib, a + ib) < 1/4$ everywhere; spurious physical-sheet poles (30) of $S_\Gamma^{\rm good}$ on $a \in (0, \pi/3), b < 0$; smoothness $\eta = 0$ across Y-junctions (Prop. 3.1)	**(a)**/**(c)**	Passed ($< 10^{-32}$)	`scripts/smx_poles_good_vs_bad.py`
N-soliton symplectic form (31), Darboux coordinates (32), and wall-crossing invariance across all $N!$ chambers ($N = 4$)	**(c)**	Passed ($< 10^{-38}$)	`scripts/smx_twisted_ring_bethe.py`
Twisted ring Bethe phase (37): exact Hirota shift $-\partial_{P_j}\Phi_{\rm ring}(j \mid l) = \Delta x_{jl}$, global sum rule (42), and covariant breather fusion (43)–(44) (Thm. 4.2)	**(c)**	Passed ($< 10^{-38}$)	`scripts/smx_twisted_ring_bethe.py`
Minimal residual ambiguity (Prop. 5.1), continuum integral crossing $J_{12}^{\rm same}(\alpha_1, \alpha_2 + \pi) + J_{12}^{\rm head}(\alpha_1, \alpha_2) = +\pi/2$ (Thm. 5.2), and small-amplitude interface laws (59)–(63)	**(c)**	Passed ($< 10^{-38}$)	`scripts/smx_residual_ambiguity_interface.py`
That any surviving candidate is the exact S-matrix of a well-defined quantum Boussinesq theory	**(d)**	Open	—